

Part XI: Fundamental Constants — Primordial Vortices and Relational Parameters

Vibrational Geometrodynamics (VG), Series I

Authors: Physicist Rodica Pop (conceptual oversight, intuition) & DeepSeek AI (systematization, formal development, and drafting)

Date: 18.01.2026

Abstract: This paper concludes the Vibrational Geometrodynamics (VG) framework by reinterpreting the fundamental constants of physics (c , G , $\hbar$, Λ). They are proposed to be emergent properties of the equilibrium state of a primary network of "quantum vortices" – nodal entities that unite a vibrational degree of freedom with an intrinsic spin. These constants are "fossils" of a cosmic phase transition triggered by the torsional instability of an initial maximum-spin configuration (the Packed Network). The model offers a path for the conceptual unification of granularity, dynamics, and geometry. This work represents the culmination of Series I of VG.

1. Introduction: From Arbitrary Values to Structural States

The fundamental constants of physics represent the inexplicable parameters of reality. VG proposes an ontological shift: these constants are not values inserted into equations but structural equilibrium properties of the dynamic fabric from which spacetime itself emerges. They are to the universe what the fundamental frequency and tension are to a vibrating membrane – properties of the substance, not of the equations.

2. The Atoms of Space: The Node-Vortex as a Quantum Oscillator with Spin

The foundation of VG is the Primary Network, whose elementary entities are not points or links, but node-vortices.

• **Definition: A node-vortex (i)** is a fundamental quantum entity with two coupled degrees of freedom:

1. Vibrational Coordinate (q_i): Describes the excitation of an internal resonance mode.

2. Intrinsic Spin ($\hat{S}_i$): Describes the orientation of an internal quantum angular momentum, the axis of the vortex.

• **Intrinsic Dynamics:** The nodal Hamiltonian is $\hat{H}_i = (p_i^2)/2 + (\omega_i^2 q_i^2)/2$, where the proper frequency ω_i is modulated by the internal spin state and relational constraints.

• **Role of Links:** The links between nodes are interaction channels through which vibrations (q) are transmitted and spins ($\hat{S}$) are compared. The coupling J_{ij} depends on the relative alignment of the vortices, $\hat{S}_i \bullet \hat{S}_j$.

In this vision, spins do not exist in space; spins constitute space. Their quantum freedom is the origin of dynamics.

3. The Primordial Mechanism: From Packed Vortices to Frozen Constants

3.1. Initial State – The Vortex Crystal: The Packed Network is a configuration where all node-vortices have perfectly aligned spins and maximum coupling. It is a state of extreme torsional tension and energy.

3.2. Torsional Trigger – The Vortex Rupture: The critical spin density creates an instability. A quantum fluctuation in the alignment of a single spin triggers a relational first-order phase transition: the alignment breaks, the torsional energy is released, and the nodal coupling rapidly decreases. This event is the inflationary epoch – the "unpacking" of the network.

3.3. Freezing of Relational Fossils: The network reaches a new relaxed dynamic equilibrium state. The properties of this state become fixed and universal:

- The propagation speed (c_{net}) of perturbations in the network → the Speed of Light (c).
- The equilibrium coupling intensity (J_c) → Newton's Gravitational Constant (G).
- The established granularity scale (ℓ_0) of the network → the Reduced Planck Constant ($\hbar$), via

$$\hbar \propto \ell_0^2 c^3 / G.$$

- The residual background tension (ν_Λ) → the Cosmological Constant (Λ).

4. Reinterpreting the Constants in the New Framework

- c : The speed of relational causality within the network.
- G : The measure of the elastic rigidity of the coupled vortex network.
- $\hbar$: The echo of area quantization emerging from the network's granularity (ℓ_0).
- Λ : The intrinsic tension of the relational "vacuum," its state of minimum energy.

5. Extension and Perspective: Torsion as an Emergent Property

The spin-based triggering mechanism leaves a collective geometric imprint. Spacetime torsion thus appears not as an added fundamental constant, but as an emergent property of the dynamics of the node-vortex network. This perspective provides a natural origin within VG for:

- The quantum spin of particles, seen as a residual excitation of the $\hat{S}_i$ degree of freedom of the node-vortices.
- A possible matter-antimatter asymmetry, which could be a consequence of the global chirality imprinted by the primordial torsional eruption.
- Einstein-Cartan relativity, whose spin-torsion coupling is reinterpreted as a dynamic emergent relation between the network's degrees of freedom, not as a primitive parameter.

6. Conclusion: A Universe of Vortices in Equilibrium

VG proposes that the search for a "theory of everything" is the search for the equilibrium conditions of a network of primitive relational entities. The fundamental constants are the answer to the question: "What are the basic properties of the equilibrium state the fabric of reality settled into after releasing its primordial tension?"

This chapter concludes Series I of Vibrational Geometrodynamics, offering a unified picture from granular-vibrational structure to cosmology. It prepares the ground for Series II (Reality, Time, and Self), where the nature of time, consciousness, and the relationship between the observer and the vibrant network will be explored based on this solid foundation.